Series & Finance Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 An amount of $5000 is invested and compounded annually at 6%. After how many years will the investment first exceed double of its initial value? 1

$FV = 5000\,(1+6\%)^n$
$10000 = 5000\,(1.06)^n$
$2 = 1.06^n$
$\ln 2 = n \times \ln(1.06)$
$n = \dfrac{\ln 2}{\ln 1.06} = 11.89$ years
$\therefore n = 12$ years

2 Emma is saving for a new car. She deposits $2200 into an annuity account at the end of every year, for four years. The account pays 3% per annum interest, compounding annually. The table shows future values of an annuity of $1. How much interest does Emma earn on her investment over the four years? 3

Future values of an annuity of $1

Years	Interest rate per annum						
	1%	1.5%	2%	2.5%	3%	3.5%	4%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0150	2.0200	2.0250	2.0300	2.0350	2.0400
3	3.0301	3.0452	3.0604	3.0756	3.0909	3.1062	3.1216
4	4.0604	4.0909	4.1216	4.1525	4.1836	4.2149	4.2465
5	5.1010	5.1523	5.2040	5.2563	5.3091	5.3625	5.5256

$4.1836 \times 2200 = \$9203.92$
Investment $= 2200 \times 4 = \$8800$
Interest $= \$9203.92 - \$8800 = \$403.92$

3 The table below shows the future value of a $1 annuity.

Future value of $1

End of year	1%	2%	3%	4%	5%	6%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500	2.0600
3	3.0301	3.0604	3.0909	3.1216	3.1525	3.1836
4	4.0604	4.1216	4.1836	4.2465	4.3101	4.3746

(i) What would be the future value of a $24 000 per year annuity at 5% per annum for 2 years, with interest compounding annually? 1

Intersection value is 2.05 (5% and 2 years)
$FV = 2.05 \times 24\,000 = \$49\,200$

(ii) An annuity of $31 800 is invested every half-year at 4% per annum, compounded six-monthly for 2 years. What is the future value of the annuity? 1

Intersection value is 4.1216 (2% and 4 years)
$FV = 4.1216 \times 31\,800 = \$131\,066.88$

(iii) Sophia aims to have a deposit for an apartment of at least $92 000 in 4 years-time by investing in an annuity. The annuity has an interest rate of 6% p.a. compounded annually. Calculate Sophia's yearly contribution to achieve the deposit. Answer to the nearest dollar. 2

Intersection value is 3.25 (5% and 4 years)
Let x be the amount to be invested
$FV = 4.3746 \times x$
$92\,000 = 4.3746 \times x$
$x = \dfrac{92\,000}{4.3746} = \$21\,030.4942\ldots \approx \$21\,030$
$\therefore \$21\,030$ is invested every year.

4 George invests $100 000 which was an inheritance, in an account which earns 4% interest, compounded annually. He intends to withdraw $$M$ at the end of each year, immediately after the interest has been paid. He wishes to be able to do this for 25 years, after which the account will be empty. What is the amount $$M$, that George will receive each year? 4

4% C.I annually

Year 1 $= 100\,000 \times 1{\cdot}04 - M$

Year 2 $= (100\,000 \times 1{\cdot}04 - M) \times 1{\cdot}04 - M$

Year 3 $= [100000 \times 1{\cdot}04^2 - M \times 1{\cdot}04 - M] \times 1{\cdot}04 - M$

$= 100\,000 \times 1{\cdot}04^3 - M \times 1{\cdot}04^2 - M \times 1{\cdot}04 - M$

$= 100\,000 \times 1{\cdot}04^3 - M(1 + 1{\cdot}04 + 1{\cdot}04^2)$

Year 25 $= 100000 \times 1{\cdot}04^{25} - M(1 + 1{\cdot}04 + \dots + 1{\cdot}04^{24})$

Now $S_n = \frac{a(r^n - 1)}{r - 1} \Rightarrow S_{25} = \frac{1(1{\cdot}04^{25} - 1)}{1{\cdot}04 - 1}$

After 25 years balance is zero

$\therefore 0 = 100000 \times 1{\cdot}04^{25} - M\frac{(1{\cdot}04^{25} - 1)}{0{\cdot}04}$

$\therefore M = \frac{100000 \times 1{\cdot}04^{25} \times 0{\cdot}04}{1{\cdot}04^{25} - 1}$

$= 6{,}401{\cdot}196\ldots = \$6{,}401{\cdot}20$

5 Stephen considers two options that will allow him to purchase a property worth $350 000 within the next fifteen years.

(i) One option is to invest an amount at the end of each year into an annuity which pays interest at 3% per annum. This table shows the future values of an annuity of $1 for periods between 10 and 16 years, for different annual interest rates. The contributions are made at the end of each year. Use the table to determine how much he would need to invest each year to achieve his target of $350 000 after 15 years. 1

Years	Interest Rate per annum					
	3%	4%	5%	6%	7%	8%
10	11.4639	12.0061	12.5779	13.1808	13.8164	14.4866
11	12.8078	13.4864	14.2068	14.9716	15.7836	16.6455
12	14.1920	15.0258	15.9171	16.8699	17.8885	18.9771
13	15.6178	16.6268	17.7130	18.8821	20.1406	21.4953
14	17.0863	18.2919	19.5986	21.0151	22.5505	24.2149
15	18.5989	20.0236	21.5786	23.2760	25.1290	27.1521
16	20.1569	21.8245	23.6575	25.6725	27.8881	30.3243

Correct future value of $1 = 18·5989

Investment required $= \frac{350000}{18{\cdot}5989}$

$= 18{,}818{\cdot}31721$

$\therefore$ Yearly investment = $18 818·32

(ii) His second option is to borrow $350 000 now at the same interest rate and pay it back in monthly repayments over 15 years. 2

Months	Interest Rate per month					
	0.15%	0.20%	0.25%	0.30%	0.35%	0.40%
176	154.5830	148.2363	142.2439	136.5834	131.2338	126.1755
177	155.3500	148.9384	142.8867	137.1719	131.7726	126.6688
178	156.1158	149.6391	143.5279	137.7586	132.3095	127.1602
179	156.8805	150.3385	144.1675	138.3436	132.8446	127.6496
180	157.6440	151.0364	144.8055	138.9268	133.3777	128.1370
181	158.4064	151.7329	145.4419	139.5083	133.9090	128.6226
182	159.1677	152.4281	146.0767	140.0880	134.4385	129.1061
183	159.9278	153.1218	146.7099	140.6660	134.9661	129.5878

This table shows the present values of an annuity of $1 for periods between 165 and 172 months, for different monthly interest rates. Use the table to find the monthly repayment necessary to pay off the loan of $350 000 in fifteen years.

Interest rate = 3% p.a = $\frac{3}{12}$ = 0·25% per month
Period = 15 years = 15 × 12 = 180 months
monthly repayment = 144·8055 for $1
Now 350000 = 144·8055 × Repayment
∴ Repayment = $\frac{350000}{144 \cdot 8055}$
= 2417·035265
∴ Repayment = $2417·04

(iii) Ba sed on your answers to (i) and (ii), determine which option would cost Stephen less each year in payments. Repayments for 1 year = 2417·04 × 12 1
= $29004·48
The investment (a) is cheaper

(iv) Outline one other factor he should consider when comparing these options.

The price of a property now will increase over 15yrs 1
Rent would be saved if he buys now

6 Steve and Stella take out a loan of $50 000 for home renovations at 3% per annum compounded monthly. They make regular repayments of $750 at the end of each month. The recurrence relation to model this situation is given by the formula $A_n = A_{n-1}(1.0025) - 750$. How much have Steve and Stella paid off the amount they borrowed, after the first three repayments? 3

$A_1 = 50000(1.0025) - 750 = 49375$
$A_2 = 49375(1.0025) - 750 = 48748.4375...$
$A_3 = 48748.44(1.0025) - 750 = 48120.31$
Reduction in balance = 50 000 − 48 120.31 = 1879.69

7 The table shows the future value of a $1 annuity for varying interest rates over different time periods. What is the present value of an annuity, correct to the nearest dollar, that would provide a future value of $34 800 after 4 years at 2% per annum, compounded half-yearly? 2

Time period	Interest rate				
	1%	2%	3%	4%	5%
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

$r = \frac{2\%}{2} = 1\%$ (per half-year) $n = 4 \times 2 = 8$
From the table, the interest factor is 8.2857.
$34800 = \text{present value} \times 8.2857$
present value = $4200

8 Jonathan's parents decide to set up an investment account to help pay for his university degree. Jonathan's parents deposit \$1000 into the account every year, which earns interest at 4% per annum, compounded yearly. They deposited the first \$1000 at the beginning of the year when Jonathan is born. Jonathan will receive the full amount at the end of the year he turns 18 years old. Let A_n be the value of the account after n years.

(i) Show that $A_3 = 1000(1.04)(1 + 1.04 + 1.04^2)$. 1

$$
\begin{aligned}
A_1 &= 1000(1.04) \\
A_2 &= (A_1 + 1000)(1.04) \\
&= 1000(1.04)^2 + 1000(1.04) \\
A_3 &= (A_2 + 1000)(1.04) \\
&= 1000(1.04)(1 + 1.04 + 1.04^2)
\end{aligned}
$$

(ii) Show that $A_{18} = 26\,000(1.04^{18} - 1)$. 2

$$
\begin{aligned}
A_{18} &= 1000(1.04)^{18} + 1000(1.04)^{17} + \ldots + 1000(1.04)^2 + 1000(1.04) \\
&= 1000(1.04)(1.04^{17} + 1.04^{16} + \ldots + 1.04 + 1)
\end{aligned}
$$

$a = 1,\ r = 1.04,\ n = 18$

$$
A_{18} = \frac{1000(1.04)(1.04^{18} - 1)}{0.04} = 26\,000(1.04^{18} - 1)
$$

(iii) The amount in the account continues to earn interest at 4% per annum. Jonathan enrols in a four-year university course, which costs $\$M$ each year, to be paid at the end of the year. As a result: $A_{19} = 26\,000(1.04^{18} - 1)(1.04) - M$. How much can Jonathan withdraw from the account each year if the entire amount is to be withdrawn over the four years? 5

$$
\begin{aligned}
A_{20} &= \left[26\,000(1.04^{18} - 1)(1.04) - M\right](1.04) - M \\
&= 26\,000(1.04^{18} - 1)(1.04)^2 - M(1 + 1.04) \\
A_{21} &= 26\,000(1.04^{18} - 1)(1.04)^3 - M(1 + 1.04 + 1.04^2) \\
A_{22} &= 26\,000(1.04^{18} - 1)(1.04)^4 - M(1 + 1.04 + 1.04^2 + 1.04^3) \\
&= 26\,000(1.04^{18} - 1)(1.04)^4 - 25M(1.04^4 - 1)
\end{aligned}
$$

When $A_{22} = 0$:

$$
26\,000(1.04^{18} - 1)(1.04)^4 - 25M(1.04^4 - 1) = 0
$$

$$
\frac{26\,000(1.04^{18} - 1)(1.04)^4}{25(1.04^4 - 1)} = M = \$7347.66
$$

Series & Finance Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Matthew invests $\$P$ at 7% per annum compounded annually. He plans to withdraw \$4000 at the end of each year for six years to cover university fees. Which of the following is the correct expression for the amount $\$A_6$ remaining in the account after the sixth withdrawal? 1

A. $A_6 = P \times 1.07^5 - 4000(1.07^6 + 1.07^5 + \cdots + 1)$

B. $A_6 = P \times 1.07^6 + 4000(1.07^5 - 1.07^4 - \cdots - 1)$

C. $A_6 = P \times 1.07^5 + 4000(1.07^6 - 1.07^5 - \cdots - 1)$

D. $A_6 = P \times 1.07^6 - 4000(1.07^5 + 1.07^4 + \cdots + 1)$

$A = P(1+r)^n = P \times (1+0.07)^1 = P(1.07)$

After 1 year $A_1 = P(1.07) - 5000$

After 2 years $A_2 = (P(1.07) - 5000) \times 1.07 - 5000$
$= P \times 1.07^2 - 5000(1.07 + 1)$

After 6 years $A_6 = P \times 1.07^6 - 5000(1.07^5 + 1.07^4 + \ldots + 1)$

2 A table of future value interest factors for an annuity of \$1 is shown. Simone deposits \$1000 into a savings account at the end of each year for 8 years. The interest rate for these 8 years is 0.75% per annum, compounded annually. After the 8th deposit, Simone stops making deposits but leaves the money in the savings account. The money in her savings account then earns interest at 1.25% per annum, compounded annually, for a further two years. Find the amount of money in Simone's savings account at the end of ten years. 2

Table of future value interest factors

Number of periods	Interest rate per period				
	0.25%	0.5%	0.75%	1%	1.25%
2	2.0025	2.0050	2.0075	2.0100	2.0125
4	4.0150	4.0301	4.0452	4.0604	4.0756
6	6.0376	6.0755	6.1136	6.1520	6.1907
8	8.0704	8.1414	8.2132	8.2857	8.3589
10	10.1133	10.2280	10.3443	10.4622	10.5817

$A = 8.2132 \times 1000 = 8213.2$

Leaving the money in her savings account for another 2 years at 1.25%,

$A = 8213.2 \times (1.0125)^2 \approx \$8419.81 (2\text{ d.p.})$.

3 Tony borrowed a personal loan of \$15 000 from the bank. The first three months of loan payments are shown in the table.

Amount borrowed	\$15 000	This table assumes the same number of days in each month. i.e. $I = P \times \frac{r}{12}$		
Annual interest rate (r)	7%			
Monthly repayment (R)	\$900			
Month (n)	Principal (P)	Interest (I)	$P+I$	$P+I-R$
1	\$15 000.00	\$87.50	\$15 087.50	\$14 187.50
2	\$14 187.50	\$82.76	\$14 270.26	\$13 370.26
3	\$13 370.26	\$77.99	\$13 448.25	\$12 548.25

(i) How much interest did Tony pay in the first three months? 1

Interest = 87.50 + 82.76 + 77.99
= \$248.25

(ii) How much of the principal would be repaid after the 4th payment? 2

Month	P	I	P + I	P + I − R
4	12548.25	73.20	12621.45	11721.45

Principal repaid = 15000 − 11721.45 = \$ 3278.55

2

(iii) Tony made a payment of \$6000 as the 4th payment. How much interest did he save in the 5th month?

Old interest $= 1172.45 \times \frac{7}{12}\% = \$68.38$

New interest $= (12621.45 - 6000) \times \frac{7}{12}\% = \$38.63$

4 (i) On the day that M Save $= \$29.75$ 2

earning 3% per annum compounded annually. On each birthday after this, her grandfather deposited \$1000 into the account, making his final deposit on Megan's 17th birthday. That is, a total of 18 deposits were made. Let A_n be the amount in the account on Megan's nth birthday, after the deposit is made. Show that $A_n = \$8554.54$.

$$A_1 = 5000(1.03) + 1000$$
$$A_2 = A_1(1.03) + 1000 = 5000(1.03)^2 + 1000(1.03) + 1000$$
$$A_3 = A_2(1.03) + 1000 = 5000(1.03)^3 + 1000(1.03)^2 + 1000(1.03) + 1000 = \$8554.54$$

(ii) On her 17th birthday, just after the final deposit is made, Megan has \$30 025.83 in her account. You are NOT required to show this. Megan then decides to leave all the money in the same account continuing to earn interest at 3% per annum compounded annually. On her 18th birthday, and on each birthday after this, Megan withdraws \$2000 from the account. How many withdrawals of \$2000 will Megan be able to make? 3

$$B_1 = A_{17}(1.03) - 2000$$
$$B_2 = A_{17}(1.03)^2 - 2000(1.03) - 2000$$
$$\vdots$$
$$B_n = A_{17}(1.03)^n - 2000[(1.03)^{n-1} + (1.03)^{n-2} + \cdots + 1.03 + 1]$$
$$= A_{17}(1.03)^n - \frac{2000(1-1.03^n)}{1-1.03} \quad \text{Solve for } B_n = 0.$$
$$A_{17}(1.03)^n - \frac{2000(1-1.03^n)}{1-1.03} = 0$$
$$A_{17}(1.03)^n - \frac{2000}{-0.03} + \frac{2000(1.03)^n}{-0.03} = 0$$
$$\left(A_{17} - \frac{2000}{0.03}\right)(1.03)^n = -\frac{2000}{0.03}$$
$$(1.03)^n = -\frac{2000}{0.03} \times \frac{1}{A_{17} - \frac{2000}{0.03}}$$
$$n = \frac{1}{\ln(1.03)} \ln\left(-\frac{2000}{0.03} \times \frac{1}{A_{17} - \frac{2000}{0.03}}\right) \approx 20.249$$

Hence, Megan can make a maximum of 20 withdrawals of \$2,000 before her account balance is insufficient.

5 Daniel wants to deposit \$500 at the start of each month into an account that earns interest at 0.4% per month. The amount in the account at the end of the nth month can be determined using the recurrence relation $A_n = 1.004(500 + A_{n-1})$, where $n = 1, 2, 3, \ldots$ and $A_0 = 0$.

(i) Use the recurrence relation to find the amount of money in the account at the end of the third month. 2

$A_1 = 1.004\,(500 + A_0)$ as $A_0 = 0$ then

$A_1 = 1.004 \times 500 = 502$

Now, $A_2 = 1.004\,(500 + A_1)$

Again $A_3 = 1.004\,(500 + A_2)$

So $A_3 = 1.004\,(500 + 1006.008) = 1512.03$

Hence, Daniel will have \$ 1512.03 to the nearest cent at the end of the third month.

(ii) Calculate the amount of interest earned in the third month. 1

At the end of the 2nd month, Daniel had \$1006.01 and at the end of the 3rd month he had \$1512.03

By subtraction, the extra amount was \$506.02, but Daniel deposited \$500, so the interest was \$6.02.

(iii) Calculate the amount of money in the account at the end of 120 months. 3

$$A_1 = 1.004 \times 500$$
$$A_2 = 1.004 \times (500 + 1.004 \times 500)$$
$$A_2 = (1.004 + 1.004^2) \times 500$$
$$A_3 = 1.004 \times (500 + 1.004 \times 500 + 1.004^2 \times 500)$$
$$A_3 = (1.004 + 1.004^2 + 1.004^3) \times 500$$
$$\vdots$$
$$A_{120} = (1.004 + 1.004^2 + \cdots + 1.004^{120}) \times 500$$

Let $K = 1.004 + 1.004^2 + \cdots + 1.004^{120}$. K is a geometric series with $a = 1.004$, $r = 1.004$ and $n = 120$.

$$K = \frac{1.004\left(1.004^{120} - 1\right)}{1.004 - 1} = 154.2464868$$

6 A fund is established to provide prizes for a basketball team's annual Awards night, $10 000 is placed in the fund one year before the first Awards night. It is decided that $450 will be withdrawn from the fund each year to purchase the annual prizes. The money in the fund is invested at 3% p.a. compounded annually with the interest paid into the fund before each annual Awards night.

(a) If A_n is the amount in dollars remaining in the fund after the n[th] Awards night, prove that $A_n = 5000(3 - 1.03^n)$ 3

$A_1 = 10000(1+0.03)^1 - 450$
$A_2 = A_1 \times (1.03) - 450$
$A_2 = 10000(1.03)^2 - 450(1.03+1)$
$A_3 = A_2 \times (1.03) - 450$
$A_3 = 10000(1.03)^3 - 450(1.03^2 + 1.03^1 + 1)$
$A_n = 10000(1.03)^n - 450(1 + 1.03 + 1.03^2 + \cdots + 1.03^{n-1})$
$A_n = 10000(1.03)^n - 450\left[\frac{1(1.03^n - 1)}{1.03 - 1}\right]$
$A_n = 10000(1.03)^n - 15000(1.03^n - 1)$
$A_n = -5000(1.03)^n + 15000$
$A_n = 5000(3 - 1.03^n)$

(b) For the fund described above, it is decided to increase the amount of money withdrawn for each Award night by 2% each year.

(i) Show that the amount remaining in the fund after the 2[nd] Awards night is $9686.50. 1

$B_1 = 10000(1.03) - 450$
$B_2 = B_1 \times (1.03) - [450 \times (1.02)]$
$B_2 = \$9686.50$

(ii) Find the amount remaining in the fund after the 25[th] Awards night. Give your answer correct to the nearest dollar. 2

$B_3 = B_2 \times (1.03) - [450 \times (1.02) \times (1.02)]$
$B_3 = [10000(1.03)^2 - 450(1.03) - 450(1.02)] \times (1.03) - 450(1.02)^2$
$B_3 = 10000(1.03)^3 - 450[(1.03)^2 + (1.02)(1.03) + (1.02)^2]$
$B_3 = 10000(1.03)^3 - 450\left[\frac{(1.03)^2\left[\left(\frac{1.02}{1.03}\right)^3 - 1\right]}{\frac{1.02}{1.03} - 1}\right]$
$B_{25} = 10000(1.03)^{25} - 450\left[\frac{(1.03)^{24}\left[\left(\frac{1.02}{1.03}\right)^{25} - 1\right]}{\frac{1.02}{1.03} - 1}\right]$
$B_{25} = \$545$

7 Lucas has opened an investment account. He will invest $*A* at the end of each month for 4 years at an interest rate of 0.5% per month, with interest compounded monthly. He wants the future value of the account to be $250 000. If the future value of $1 is $54.0978, find the value of *A* correct to two decimal places. 1

$A \times 54.0978 = 250000$ so $A \approx 4621.26.$

8 Mr and Mrs Lam each decide to invest some money each year to help pay for their daughter's first car. The parents choose different investment strategies.

(i) Mr Lam makes 18 yearly contributions of \$2000 into an investment fund. He makes his first contribution on the day his daughter is born, and his final contribution on her seventeenth birthday. His investment earns 5% compound interest per annum. Find the total value of Mr Lam's investment on his daughter's eighteenth birthday. 2

$$A_{18} = 2000\times 1.05 + 2000\times 1.05^2 + 2000\times 1.05^3 + \ldots + 2000\times 1.05^{18}$$
$$A^{18} = \frac{2000\times 1.05\left(1.05^{18}-1\right)}{1.05-1} = \$59078.007\ldots \approx \$59078.01$$

(ii) Mrs Lam makes her contributions into another fund. She contributes \$2000 on the day of her daughter's birth, and increases her annual contribution by 5% each year. Her investment also earns 5% compound interest per annum. Find the total value of Mrs Lam's investment on her daughter's third birthday (just before she makes her fourth contribution). 1

$$A_0 = 2000 \qquad A_1 = 2000\times 1.05$$
$$A_2 = \left(2000\times 1.05 + 2000\times 1.05\right)\times 1.05 = 2\times 2000\times 1.05^2$$
$$A_3 = \left(2\times 2000\times 1.05^2 + 2000\times 1.05^2\right)\times 1.05 = \$6615$$

(iii) Mrs Lam also makes her final contribution on her daughter's seventeenth birthday. Find the total value of Mrs Lam's investment on her daughter's eighteenth birthday. 1

$$A_{18} = 18\times 2000\times 1.05^{18} = \$86638.292\ldots \approx \$86638.29$$